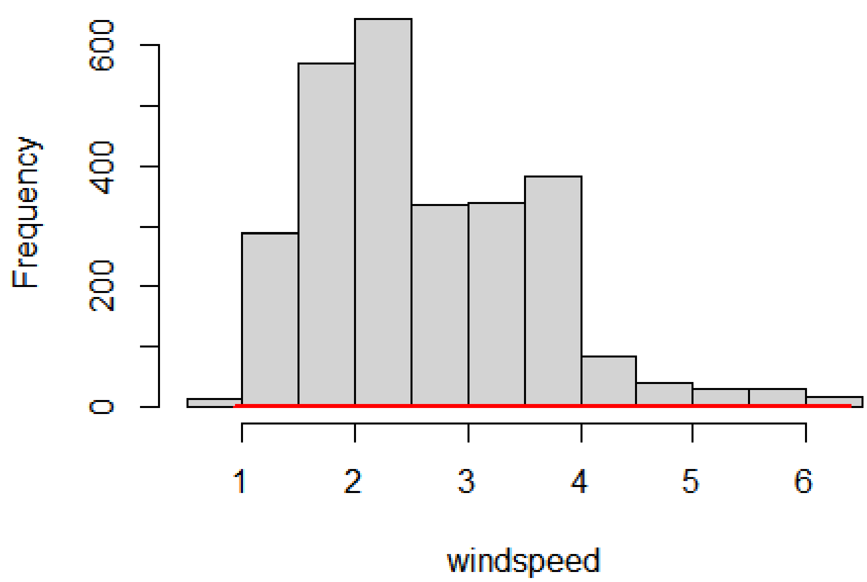
ACTU_302_10967696.R

2024-07-18

```
### (a) reading the dataset into r
#install.packages("openxlsx")
library(openxlsx)
data <- read.xlsx("Raw dataset.xlsx")


### Displaying few observations
head(data)

##   Code Group   Family        Species       P50        KS    AL/.AS
D
## 1  510   Gym Pinaceae     Abies alba -3.710000        NA        NA 20.4276
1
## 2  622   Gym Pinaceae Abies balsamea        NA 0.2200855 0.2136752 20.0000
0
## 3  965   Gym Pinaceae Abies balsamea -1.995803 3.2118270        NA 28.9751
8
## 4 2055   Gym Pinaceae Abies concolor -4.831500        NA        NA       N
A
## 5 2540   Gym Pinaceae    Abies fabri        NA        NA 0.6330156       N
A
## 6 2560   Gym Pinaceae    Abies fabri        NA        NA 0.4769127       N
A
##          WD Height        μ       MAT  MAP        VPD     MI        TS
## 1 0.4581269   30.0 3.283333 9.5506667  707  0.5621208 0.9201  581.9456
## 2        NA    2.0 2.700000 7.4136667 1078  0.5221427 1.0586 1001.7569
## 3 0.4467858    1.5 3.250000 2.6071667  486  0.3867455 0.6551 1095.9092
## 4        NA     NA 3.425000 6.0613333  533  1.0478634 0.6447  818.5885
## 5        NA     NA 2.091667 3.6983334  847 15.3554876 0.9333  618.9011
## 6        NA     NA 2.300000 0.7773333  828 15.3554876 0.9958  691.6901
##          PS    MTCM PDM Latitude Longitude
## 1 27.566586  -3.024  37 45°43' N    3°3' E
## 2  9.327823 -13.848  70 42°53' N  72°19' W
## 3 68.913518 -20.428  15 53°25' N 113°45' W
## 4 19.860021 -11.524  29 40°48' N 111°46' W
## 5 93.378892  -9.944   5  29°25'N  101°55'E
## 6 92.748188 -13.508   4  29°22'N  101°55'E
##                                            References
## 1           Cochard, 2006, Comptes Rendus Physique
## 2 Ewers & Zimmermann, 1984a, Physiologia Plantarum
## 3            Hacke & Jansen, 2009, New Phytologist
## 4            Sperry & Ikeda, 1997, Tree Physiology
## 5                        Xu, 2020, New Phytologist
```

```
## 6                         Xu, 2020, New Phytologist
##                         Location Country
## 1               Clermont-Ferrand  France
## 2 Petersham, Massachusetts (MA)     USA
## 3   25 km southwest of Edmonton  Canada
## 4                          Utah     USA
## 5               Sichuan province   China
## 6               Sichuan province   China

windspeed <- data$µ
head(windspeed)

## [1] 3.283333 2.700000 3.250000 3.425000 2.091667 2.300000

### Descriptive statistics
#install.packages("psych")
library(psych)
describe(windspeed)

##    vars    n mean   sd median trimmed  mad  min max range skew kurtosis
se
## X1    1 2780 2.64 1.01   2.39    2.56 0.99 0.93 6.4  5.48 0.84     0.72 0.
02

hist(windspeed,
     main = "")


### (b) loglikelihood function
log_likelihood <- function(x,parameters)
{
  alpha <- parameters[1];s <- parameters[2]
  survival <- exp(-((x/s)^alpha))
  pdf <- (alpha*x^(alpha-1)/(s^alpha))*(exp(-((x/s)^alpha)))
  log_lik <- sum(log(pdf))
  return(-log_lik)
}



### (c) Fiting the Windspeed data to the Weibul distribution
### Taking off the NA responses from the windspeed
windspeed <- windspeed[!is.na(windspeed)]
mle <- optim(par = c(2,3),
             fn = log_likelihood,
             method = "Nelder-Mead",
             x = windspeed);mle

## $par
## [1] 2.766691 2.964861
##
```

```
## $value
## [1] 3900.877
##
## $counts
## function gradient
##       49       NA
##
## $convergence
## [1] 0
##
## $message
## NULL

### (d)
alpha <- mle$par[1];s <- mle$par[2]
exceed_prob <- exp(-((6/s)^alpha))
emp_exceed_prob <- length(windspeed[windspeed>6])/length(windspeed)


### For normal
lines(x = sort(windspeed),
      pnorm(q = sort(windspeed),mean = mean(windspeed),sd = sqrt(var(windspee
d))),
      lwd = 2,
      col = "red")
```



```
### for Weibil
density <- 1 - exp(-((sort(windspeed)/s)^alpha))
#lines(x = sort(windspeed,
#                y = density))
```